

8-26 ALTAM Additional practice solutions

ALTAM Sample Questions

Question 28

The annual probability transition matrix for an insured age $60+k, k=0,1,\dots,9$ is:

$$\begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{pmatrix} 0.90-0.01k & 0.05 & 0.05+0.01k \\ 0.70-0.01k & 0.20 & 0.10+0.01k \\ 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

Calculate ${}_2p_{60}^{01}$.

$$\begin{array}{lcl} 0 \rightarrow 0 \rightarrow 1 & (0.90)(0.05) & = 0.045 \\ 0 \rightarrow 1 \rightarrow 1 & (0.05)(0.2) & = 0.01 \\ & \hline & 0.055 \end{array}$$

Graders' Comments:

1. This part was done correctly by almost all candidates.
2. For those who did not receive full credit, two common errors were to use an incorrect annual probability of transition and to calculate only one of two terms.

Spring 2023 Exam ALTAM

1.

(10 points) You are given the following information about a portfolio of in-force policies.

- Policyholders are either Healthy (State 0), Temporarily Disabled (State 1), Permanently Disabled (State 2), or Dead (State 3).
- At time 0, there are 14 Healthy lives, 0 Temporarily Disabled lives, and 1 Permanently Disabled life.
- All 15 lives are independent.
- Transitions are modeled as a homogeneous Markov model.
- The annual transition matrix is

$$P = \begin{pmatrix} 0.80 & 0.10 & 0.05 & 0.05 \\ 0.20 & 0.50 & 0.10 & 0.20 \\ 0 & 0 & 0.70 & 0.30 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- The two-year transition matrix, but with some missing values denoted by --, is

$${}_2P = \begin{pmatrix} 0.660 & -- & 0.085 & -- \\ 0.260 & 0.270 & 0.130 & -- \\ 0 & 0 & -- & -- \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- (2 points) Calculate the missing values in the two-year transition matrix.

$$\begin{array}{lcl} {}_2p_x^{01} & 0 \rightarrow 0 \rightarrow 1 & (0.8)(0.1) = 0.08 \\ & 0 \rightarrow 1 \rightarrow 1 & (0.1)(0.5) = 0.05 \\ & & \hline & & 0.13 \end{array}$$

$$\begin{array}{lcl} {}_2p_x^{03} & 0 \rightarrow 0 \rightarrow 3 & (0.8)(0.05) = 0.04 \\ & 0 \rightarrow 1 \rightarrow 3 & (0.1)(0.2) = 0.02 \\ & 0 \rightarrow 2 \rightarrow 3 & (0.05)(0.3) = 0.015 \\ & 0 \rightarrow 3 \rightarrow 3 & (0.05)(1) = 0.05 \\ & & \hline & & 0.125 \end{array}$$

$$0.66 + 0.13 + 0.085 + 0.125 = 1 \quad \checkmark$$

$$\begin{array}{ll}
 2P_X^{13} & 1 \rightarrow 1 \rightarrow 3 \quad (0.5)(0.2) = 0.1 \\
 & 1 \rightarrow 0 \rightarrow 3 \quad (0.2)(0.05) = 0.01 \\
 & 1 \rightarrow 2 \rightarrow 3 \quad (0.1)(0.3) = 0.03 \\
 & 1 \rightarrow 3 \rightarrow 3 \quad (0.2)(1) = 0.20 \\
 & \quad \quad \quad \underline{0.34}
 \end{array}$$

$$0.26 + 0.27 + 0.13 + 0.34 = 1 \checkmark$$

$$2P_X^{22} \quad 2 \rightarrow 2 \rightarrow 2 \quad (0.7)(0.7) = 0.49$$

$$\begin{array}{ll}
 2P_X^{23} & 2 \rightarrow 2 \rightarrow 3 \quad (0.7)(0.3) = 0.21 \\
 & 2 \rightarrow 3 \rightarrow 3 \quad (0.3)(1) = 0.3 \\
 & \quad \quad \quad \underline{0.51}
 \end{array}$$

$$0.49 + 0.51 = 1 \checkmark$$

Examiners' Comments:

(a) Candidates did well on this part. The most common mistake was calculation error.